



PSDL Final practical Program SPPU SEIT-4th sem

Information technology (Savitribai Phule Pune University)



Scan to open on Studocu

Addition of Array

```
#include <pic18f4550.h>
```

```
void main(void)
```

```
{
```

```
    int i, n=10, sum=0;
```

```
    TRISB=0;
```

```
    int number[]={1,2,3,4,5,6,7,8,9,10};
```

```
    for(i=0; i<n; i++)
```

```
    {
```

```
        sum = sum + number[i]; // or sum += number[i];
```

```
    }
```

```
    PORTB = sum;
```

```
    return;
```

```
}
```

Addition of Numbers

```
# include <PIC18F4550.h>

void main (void)
{
    unsigned int num1 , num2 ,sum ;

    TRISB =0 ;

    //LATB =0 ;

    num1 = 0x04;

    num2 = 0x05;

    sum = num1 + num2 ;

    PORTB = sum ;

    PORTC = num1 ;

    PORTD = num2 ;

    return ;

}
```

Multiplication of Number

```
# include <PIC18F4550.h>
```

```
void main (void)
```

```
{
```

```
unsigned int num1 , num2 ,sum ;
```

```
TRISB =0 ;
```

```
//LATB =1 ;
```

```
num1 = 0x04;
```

```
num2 = 0x05;
```

```
sum = num1 * num2 ;
```

```
PORTB = sum ;
```

```
PORTC = num1 ;
```

```
PORTD = num2 ;
```

```
return ;
```

```
}
```

LED BLINKING

// LED Blinking

#include <PIC18f4550.h>

void delay(unsigned int time)

{

unsigned int i,j;

for(i=0;i<time;i++)

for(j=0;j<5000;j++);

}

void main(void)

{

TRISB = 0x00;

LATB = 0xFF;

while(1)

{

LATB = ~LATB;

delay(200);

}

}

Buzzer

```
#include <PIC18F4550.h>

int c=0;

void __interrupt() tmr1_isr(){

c++;

if(c==1000){

c=0;

LATAbits.LATA5=~LATAbits.LATA5;

}

TMR1IF=0;

TMR1=53536;

}

void main(void) {

T1CON = 0x80;

TRISAbits.TRISA5=0;

LATAbits.LATA5=0;

TMR1IE=1;

TMR1IF=0;

GIE=1;

PEIE=1;

TMR1=53536;

TMR1ON=1;

while(1);

return;
```

}

LCD interfacing

```
#include <xc.h>
```

```
#define LCD_EN LATAbits.LA1
```

```
#define LCD_RS LATAbits.LA0
```

```
#define LCDPORT LATB
```

```
void lcd_delay(unsigned int time)
```

```
{
```

```
    unsigned int i , j ;
```

```
        for(i = 0; i < time; i++)
```

```
        {
```

```
            for(j=0;j<100;j++);
```

```
        }
```

```
}
```

```
void SendInstruction(unsigned char command)
```

```
{
```

```
    LCD_RS = 0;          // RS low : Instruction
```

```
    LCDPORT = command;
```

```
    LCD_EN = 1;          // EN High
```



```

    lcd_delay(10);

    LCD_EN = 0;          // EN Low; command sampled at EN falling edge

    lcd_delay(10);
}

```

```

void SendData(unsigned char lcddata)

```

```

{
    LCD_RS = 1;          // RS HIGH : DATA

    LCDPORT = lcddata;

    LCD_EN = 1;          // EN High

    lcd_delay(10);

    LCD_EN = 0;          // EN Low; data sampled at EN falling edge

    lcd_delay(10);
}

```

```

void InitLCD(void)

```

```

{
    ADCON1 = 0x0F;

    TRISB = 0x00; //set data port as output

    TRISAbits.RA0 = 0; //RS pin

    TRISAbits.RA1 = 0; // EN pin


    SendInstruction(0x38); //8 bit mode, 2 line,5x7 dots

    SendInstruction(0x06); // entry mode

    SendInstruction(0x0C); //Display ON cursor OFF
}

```

```

    SendInstruction(0x01);    //Clear display

    SendInstruction(0x80);    //set address to 1st line

}

/
*****
*****/

unsigned char *String1 = "Vinayak";

unsigned char *String2 = "SEIT-71";


void main(void)
{
    ADCON1 = 0x0F;

    TRISB = 0x00;    //set data port as output

    TRISAbits.RA0 = 0; //RS pin

    TRISAbits.RA1 = 0; // EN pin


    SendInstruction(0x38);    //8 bit mode, 2 line,5x7 dots

    SendInstruction(0x06);    // entry mode

    SendInstruction(0x0C);    //Display ON cursor OFF

    SendInstruction(0x01);    //Clear display

    SendInstruction(0x80);    //set address to 1st line


    while(*String1)

    {

```

```
SendData(*String1);  
String1++;  
}
```

```
SendInstruction(0xC0); //set address to 2nd line  
while(*String2)  
{  
    SendData(*String2);  
    String2++;  
}
```

```
while(1);  
  
}
```

Tempreature

```
#include <pic18f4550.h>

#include<stdio.h>

#define LCD_EN LATAbits.LA1

#define LCD_RS LATAbits.LA0

#define LCDPORT LATB
```

```
unsigned char str[16];
```

```
void lcd_delay(unsigned int time)
```

```
{
    unsigned int i, j;
    for (i = 0; i < time; i++)
    {
        for (j = 0; j < 100; j++);
    }
}
```

```
void SendInstruction(unsigned char command)
```

```
{
    LCD_RS = 0;

    LCDPORT = command;

    LCD_EN = 1;

    lcd_delay(10);
```

```
LCD_EN = 0;

lcd_delay(10);

}
```

```
void SendData(unsigned char lcddata)
```

```
{

    LCD_RS = 1;

    LCDPORT = lcddata;

    LCD_EN = 1;

    lcd_delay(10);

    LCD_EN = 0;

    lcd_delay(10);

}
```

```
void InitLCD(void)
```

```
{

    ADCON1 = 0X0F;

    TRISB = 0X00;

    TRISAbits.RA0 = 0;

    TRISAbits.RA1 = 0;

    SendInstruction(0x38);

    SendInstruction(0x06);

    SendInstruction(0x0C);

    SendInstruction(0x01);

    SendInstruction(0x80);

}
```

```
}
```

```
void LCD_display(unsigned int row, unsigned int pos, unsigned char *ch)
```

```
{
```

```
    if (row == 1)
```

```
        SendInstruction(0x80 | (pos - 1));
```

```
    else
```

```
        SendInstruction(0xC0 | (pos - 1));
```

```
    while (*ch)
```

```
        SendData(*ch++);
```

```
}
```

```
void ADCInit(void)
```

```
{
```

```
    TRISEbits.RE2 = 1;
```

```
    ADCON1 = 0b00000111;
```

```
    ADCON2 = 0b10101110;
```

```
}
```

```
unsigned short Read_Temp(void)
```

```
{
```

```
    ADCON0 = 0b00011101;
```

```
    GODONE = 1;
```

```
    while (GO_DONE == 1);
```

```
    return ADRES;
```

```
}
```

```
int main(void)
```

```
{
```

```
    unsigned int temp;
```

```
    InitLCD();
```

```
    ADCInit();
```

```
    LCD_display(1, 1, "Temperature:");
```

```
    while (1)
```

```
    {
```

```
        temp = Read_Temp();
```

```
        temp = ((temp * 500) / 1023);
```

```
        printf(str, "%d'C ", temp);
```

```
        LCD_display(2, 1, str);
```

```
        lcd_delay(9000);
```

```
    }
```

```
    return 0;
```

```
}
```

7 segment

```
#include <pic18f4550.h>

#define ONE    0b0000110
#define TWO    0b1011011
#define THREE  0b1001111
#define FOUR   0b1100110
#define FIVE   0b1101101
#define SIX    0b1111101
#define SEVEN  0b0000111
#define EIGHT  0b1111111
#define NINE   0b1101111
#define ZERO   0b0111111

#define SEG1   LATAbits.LATA0
#define SEG2   LATAbits.LATA1

const unsigned char segData[10] = {ZERO, ONE, TWO, THREE, FOUR, FIVE, SIX,SEVEN, EIGHT, NINE};

void delay(unsigned int time)
{
    int i;
    while(time--)
        for(i=0;i<500;i++);
}
```



```

int main(void)
{
    unsigned char count = 0;
    unsigned char digit1, digit2;
    unsigned int i;

    TRISB = 0x00;

    LATA = 0x00;

    TRISAbits.TRISA0 = 0;
    TRISAbits.TRISA1 = 0;

    while(1)
    {
        for(count=0;count<100;count++)
        {
            digit1 = count / 10;
            digit2 = count % 10;
            for(i=0;i<100;i++)
            {
                SEG1 = 1;
                SEG2 = 0;
                LATB = ~segData[digit1];
                delay(2);
                SEG1 = 0;
                SEG2 = 1;
                LATB = ~segData[digit2];
            }
        }
    }
}

```

```
        delay(2);  
    }  
}  
}  
return 0;  
}
```